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SOAP classifier for free-text clinical notes with domain-specific pre-trained language models

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ABSTRACT

The increasing use of electronic health records (EHRs) in healthcare has led to a significant amount of unstructured clinical text data. This paper proposes a model for classifying free-text clinical notes in the Portuguese language in sentences following the SOAP (Subjective, Objective, Assessment, and Plan) note standard using domain-specific pre-trained language models. Among the five pre-trained BERT models tested, BioBERTptRT achieved the best results with a precision of 0.9461, accuracy of 0.9434, recall of 0.9437, and F1-score of 0.9435. BioBERTptRT, specialized in the Portuguese language, clinical terminology, and the medical group's domain, outperformed the other models, showing a 0.28% increase in the F1-score compared to the second most specialized model. The proposed model focuses on high-level sentence classification rather than entity-level classification and aims to structure clinical notes at a broader level. The study utilizes a private database of 10,000 anonymized health records containing 234,673 clinical notes. These notes were divided into 1,183,345 unique sentences used to retrain BioBERTptRT. Additionally, 100,021 sentences were manually labeled for use in fine-tuning the models This work contributes to the structuring of clinical notes by showcasing the performance improvements obtained through domain specialization in BERT networks. Additionally, it presents an analysis of the performance gains achieved by BioBERTpt compared to mBERT, both in our study and other related works. Furthermore, this study provides a valuable comparison between the distribution of sentences and the results obtained in this research and similar studies conducted in English.

1. Introduction

Information and communication technology has revolutionized healthcare globally, with electronic health records (EHR) playing a significant role (Roehrs, Da Costa, da Rosa Righi, & De Oliveira, 2017). One of the features of the EHR is to store the patient's clinical data, but despite attempts to structure this information, important details are still recorded in free text (Almeida & Matos, 2020). Up to 80% of this medical information is unstructured data written in natural language (Maximilian, J., & Michael, 2019; Tognola, Murri, & Cuda, 2018). Among the stored information, one particular type stands out: free-text clinical notes. These notes hold valuable information regarding a patient's medical history and ongoing treatments. They are shared among doctors from various specialties and institutions during both the creation and consultation processes (Ross, Schilling, Fernald, Davidson, & West, 2010). The structured registration of this information in the health record by healthcare providers makes it easier to access, reuse, and share. However, filling and adopting structured data in the health record also presents challenges (Rosenbloom et al., 2011).

In this context, numerous studies have arisen with a focus on structuring this information. These studies analyze this free-text and structure its information in diverse ways. Primarily, these studies are predominantly conducted in the English language (Fu et al., 2020; Kersloot, van Putten, Abu-Hanna, Cornet, & Arts, 2020; de Oliveira, da Costa, & Antunes, 2021).

This paper presents the results of our model for structuring clinical notes in Portuguese by classifying each sentence into one of the categories of SOAP notes (Subjective, Objective, Assessment, and Plan). The focus is on a high-level structure that classifies notes at the sentence level rather than the entity level and its relationships.

More than 200 million people speak the Portuguese language on all continents. Of the Neo-Latin languages, it was the first to expand outside the European continent. Portuguese is the official language of seven countries (Cegalla, 2008).

SOAP notes are structured as problem-oriented to standardize patient care documentation and improve communication between healthcare professionals. The method is widely used by healthcare providers,

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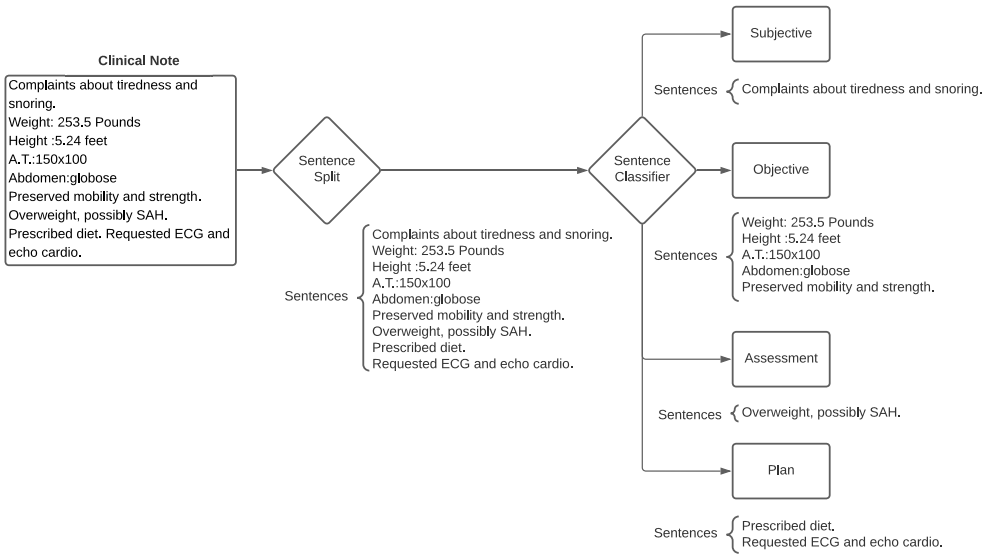


Fig. 1. Model proposed for sentence classification.

including pharmacists, to share information about patient care (Lisenby et al., 2018). Each standard category provides context about the note, guaranteeing a better understanding. For example, knowing the context in which a medication is described, such as "medications: Prilosec 20 mg 1 tablet a day" or "Prescribed Prilosec 20 mg" is crucial in understanding whether the medication is currently being taken or was prescribed during the visit (Mowery, Wiebe, Visweswaran, Harkema, & Chapman, 2012).

In this way, we modeled the problem as a sentence classification task of NLP. More specifically, a multiclass and single-label sentence classification task where the free-text clinical notes are divided into sentences and classified, receiving one of the four classes of SOAP notes. Fig. 1 shows the proposed model.

As a dataset for training and evaluation, we used a private database containing 10,000 health records, already anonymized, with 234,673 clinical notes divided into 1,183,345 single sentences. Of these single sentences, 100,021 were labeled manually (gold standard) and used to train and evaluate the models.

Various tools, proposals, and models are available to address NLP challenges in the medical field (Kim, Nam, Wang, & Zhu, 2020), but most of them concentrate on the English language. Meanwhile, works that operate out of the English language present more significant challenges precisely because of this lack of tools and previous works (de Oliveira et al., 2021). Considering this, we have chosen a BERT model because it currently represents state of the art in most of the NLP challenges (Devlin, Chang, Lee, & Toutanova, 2019; Qin et al., 2021), including tasks in the field of medicine (Alsentzer et al., 2019; Xie, Bishop, Tiwari, & Ananiadou, 2022).

Another characteristic of BERT models is the ability to be retrained to adapt to a specific domain, such as biomedical or legal, using domain-specific training data. Given the availability of pre-trained models for both Portuguese language and biomedical domains, this study examined five pre-trained network models, encompassing four existing BERT models: Multilingual BERT (Portuguese domain), DISTILBERT (Portuguese domain), BioBERT (biomedical domain), BioBERTpt (Portuguese and biomedical domain). Additionally, a pre-trained model was created by retrained BioBERTpt using our dataset's 234,673 clinical notes. The primary objective of this latter model is to assess the performance improvement of the BERT model when retrained within a more specific domain, such as a medical group or hospital network.

The SKCV (Stratified K-Fold Cross-Validation) technique was applied with N-5 to evaluate the models. As for performance metrics, the following were used: accuracy, precision, recall, and F1-score. In this way, the main contributions of this work are:

- High-level structuring model for sentence classification contained in free-text clinical notes of electronic health records in the Portuguese language following the SOAP note standard;
- Analysis of performance gain with domain specialization in BERT models;
- Comparison of the distribution of sentences and results of our work and similar work in English.

The article is divided as follows. Section 2 presents the Background. Section 3 introduces the related work. Section 4 describes the materials and methods. In Section 5, we present the results and discussion. Finally, Section 6 highlights the main conclusions and directions for future work.

2. Background

2.1. SOAP

The SOAP note is the primary way for medical practitioners to document and keep track of a patient's status, diagnosis, and treatment in inpatient and outpatient clinics. SOAP notes were designed to be a standardized method, reorganizing the structure of the medical record from simply documented physician notes to those structured around a well-defined list of a patient's medical problems (Alvarez & Reinhart, 2020; Cameron & Turtle-Song, 2002). It consists of four classes, each representing a different aspect of the medical record structure. The Subjective class covers patient complaints, symptoms, and patient history data. The Objective class encompasses data from physical examination, monitoring, and any investigations. The Assessment class evaluates the patient's condition, medical problems, needs, and prognosis, while the Plan class outlines current and future plans for investigations, treatments, education, or any other future actions (Ilan et al., 2012).

2.2. Sentence classification

Sentence classification in NLP involves two steps. The first step involves dividing the document into sentences using special characters (e.g., punctuation symbols, quotation marks, parentheses, brackets) or advanced techniques such as conditional random fields (Tomanek, Wermter, & Hahn, 2007). In the second step, each sentence is labeled with one class (single-label), as in our proposal, or more labels (multilabel) (Wang, Hu, Li and Yu, 2021).

2.3. BERT and pre-trained network models

The BERT is a context-aware language model with state-of-the-art performance in various NLP tasks (Alsentzer et al., 2019). It is projected to pre-train bidirectional representations of unlabeled text, considering both the right and left context in all layers. This makes it possible to fine-tune the BERT pre-trained model with just one additional output layer, thereby creating models for a wide range of tasks, such as answering questions and language inferences, without substantial modifications to the specific task architecture (Devlin et al., 2019). Training a BERT model involves two steps: pre-training and fine-tuning. During pre-training, the model is trained on unannotated data, which can be a large corpus of generic texts such as BooksCorpus (800 million words) and Wikipedia in English (2.5 billion words) (Lee et al., 2019). In fine-tuning, the BERT model is initialized with pre-trained parameters, and the network is trained on labeled data to perform a specific task. An additional benefit of BERT is that it can be pre-training on a pre-trained BERT model. An example is SciBERT (Beltagy, Lo, & Cohan, 2019), a pre-trained language model specifically geared towards scientific text. It was created from the English generic version of a pre-trained BERT network and incorporates the complete text of 1.14 million biomedical and computer science papers from the Semantic Scholar corpus (Ammar et al., 2018). BERT models have demonstrated superior performance compared to other methods in recognition of medical entities (Schneider et al., 2020; Symeonidou, Sazonau, & Groth, 2019).

For training and evaluating the proposed model, four existent models of pre-trained BERT networks were chosen: Multilingual BERT (mBERT), DistilBERT (base-pt), BioBERT, and BioBERTpt.

Multilingual BERT (mBERT) is a pre-trained language model trained in 104 languages and has demonstrated impressive cross-lingual capabilities on various NLP tasks even without relying on explicit cross-lingual signals. To balance the representation of different languages, mBERT employs a training strategy that increases the frequency of low-resource language words and decreases that of high-resource language words (Devlin et al., 2019; Wu & Dredze, 2020).

DistilBERT (Distilled BERT) is a smaller, faster, general-purpose, pre-trained version of BERT that maintains nearly the same language understanding abilities. It is created using knowledge distillation, a technique where a compact model is trained to imitate the performance of a larger model or multiple models, resulting in models that are lighter and faster to run during inference and requiring less computational resources during training (Santos, Dutra, Parreiras, & Brandão, 2021). DistilBERT preserves 97% of BERT's language understanding with a 60% reduction in model size, running it 60% faster during inference. The smaller model size also makes it ideal for fine-tuning various downstream tasks, as it is flexible like larger models but compact enough to run on edge devices like mobile phones (Sanh, Debut, Chaumond, & Wolf, 2019).

BioBERT is a specialized language model for representing the biomedical domain in English. It was created from the generic version of a pre-trained BERT network, incorporating the PubMed Abstracts corpus (4.5 billion words) and PMC Full-text articles (13.5 billion words) into its training (Lee et al., 2019).

BioBERTpt (Schneider et al., 2020) was specialized for the biomedical domain in Portuguese from the BERT multilingual, incorporating more clinical notes from Brazilian hospitals (27,7 million words) and the part in Portuguese of the Scielo and Pubmed databases (16,4 million words).

Similar to the performance improvement observed in BioBERTpt compared to mBERT and BioBERT, there is a possibility of achieving even better results by further customizing pre-training for a specific biomedical domain, such as a specific medical group or hospital network. To explore this hypothesis, we conducted a retraining process on BioBERTpt using our proprietary dataset, which comprised 1,183,345 unique sentences containing approximately 1.6 million words. The resulting model from this retraining was named BioBERTptRT.

Table 1 presents a comparison of the domains and features of each BERT model used in work.

Table 1
Comparison of the domains and characteristics of BERT models. The local Domain represents specialization for a specific medical group or hospital network.

Model	Lighter/ Faster	Portuguese domain	Biomedical domain	Local domain
mBERT		X		
DistilBERT	X	X		
BioBERT			X	
BioBERTpt		X	X	
BioBERTptRT		X	X	X

3. Related work

Given the limited number of studies specifically addressing SOAP sentence classification, this section incorporates those works and includes relevant research on sentence classification in the medical field and NLP tasks related to free-text clinical notes in the Portuguese language. By encompassing these additional works, we aim to provide a comprehensive overview of the topic.

Mowery et al. (2012) built an automated SOAP classifier for emergency department reports, having as a dataset 50 emergency department reports composed of 4130 sentences that were manually classified (multilabel/multiclass). The study applied several models to solve the problem and has had the best performance with support vector machine (SVM) classifiers with F1-scores of 93.9 (Subjective), 94.5 (Objective), 75.7 (Assessment), and 77.0 (Plan).

Kenei, Moso, Omullo, and Oboko (2019) present a deep learning model to identify important medical sentences in patients' medical charts and classify them into categories such as symptoms, diagnosis, disease, treatment, and personal identification information. The study includes an equivalent table of its terms with the SOAP note. As a solution, it is proposed to use convolutional residual with sentence embeddings and range normalization for classification having the raw medical text charts as input of the model. However, it is a work in progress and does not present a result or evaluation.

Analyzing broader studies on sentence classification in the medical area, Zhang, Chu, Chen, and Shang (2018) classified the sentences into health status, hospital diagnosis, treatment process, medical treatment, and clinical monitoring return appointment and created a second classification layer extracting the sentence temporality, in other words, when the sentence occurred to structure the information in a timeline.

Wu et al. (2020) classified sentences, identified entities, and related them in a study on kidney disease pathological examination reports to generate structured forms based on a medical dictionary. Still, the result of the study was only a rewritten sentence. Three studies (Moen et al., 2020, 2019, 2018) presented different approaches for the same sentence classification problem suggesting subject headings for individual sentences in nursing notes.

Although not finding studies for sentence classification in the Portuguese language are similar to the one proposed in this work, other works on NPL tasks in free-text clinical notes in Portuguese can be cited: Schneider et al. (2020) in the work that presents BioBERTpt, it also presents, as a way of evaluating its model, a NER (Named Entity Recognition) tool for medical entities with a precision of 0.917, recall 0.935, and F1-score 0.926. Souza et al. (2021) propose a multilabel approach to clinical named entity recognition in Portuguese. They use, as a dataset, SemClinBr (Silva e Oliveira et al., 2020), a semantically annotated corpus for the Brazilian Portuguese clinical NLP tasks containing 1000 clinical notes from Nephrology, Cardiology, and Endocrinology areas, with 10,000 sentences and 16,315 unique tokens. Their best results in the tested models were the BioBERTpt, with a precision of 56.6%, recall 55.7%, and f1-score 56.1%. Reys et al. (2020) predict multiple ICD-10 (International Classification of Diseases) codes from Brazilian Portuguese clinical notes, using the HSL dataset, containing de-identified documents linked to patients from the Syrian-Lebanese Hospital (HSL) with a total of 77,005 admissions from 51,298

Table 2
Distribution of the medical specialties of clinical notes of the dataset used in the work.

Specialties	Clinical notes
General clinic	133 973
Gynecology	20 848
Dermatology	14 453
Psychiatry	12 707
Ophthalmology	12 530
Cardiology	12 508
Nutritionist	10 468
Urology	6570
Psychology	5633
Endocrinology	4983

single patients. Their best results were a convolutional neural network with attention mechanisms (CNN-Att) to predict diagnosis ICD codes, with a precision of 54.3%, recall 43.8%, and f1-score 48.5%.

4. Materials and methods

This section presents the dataset used in the work, how it was divided into sentences, the labeling process, training parameters, and metrics used to evaluate the model.

4.1. Dataset

The dataset used in this study is private and provided by a medium-sized medical group located in the southern region of Brazil. This group consists of 5 clinics that provide private or affiliated health-care services. The dataset comprises 10,000 randomly preselected medical records from patients registered between May 18, 2010, and April 14, 2020. The patient's and doctor's identification were already anonymized when the medical records were delivered. This work has had the approval of the Unisinos Research Ethics Committee under the CAAE (Certificate of Presentation for Ethical Appreciation) number 61086722.0.0000.5344.

The medical records comprise two unstructured textual components: the patient's permanent record and a series of entries called clinical notes. Each medical record has only one permanent record. It is edited by multiple doctors, with no history or date, containing mainly information of little relevance or duplicated from the medical notes. On the other hand, each clinical note represents one single clinical service performed by one of the 83 accredited doctors in one of the ten specialties provided by the institution. Table 2 shows the division of the clinical notes by specialty. The database has 234,673 clinical notes, with an average of 23.4673 clinical notes per patient. For this study, we chose to work only with the patient's clinical notes since it contains much more information and has greater relevance.

4.2. Preprocessing and labeling

The first data processing applied to the dataset was the division of clinical notes into sentences. Each clinical note was divided into a sentence list. The following rules were used for the division: newline ("\n") or a period followed by a space and capital character representing the beginning of a new sentence. It generated 2,979,987 sentences, an average of approximately 12.69 sentences per clinical note of medical record and 297 sentences per patient.

After the division of the sentences, two aspects drew attention: the number of short sentences and the number of duplicate sentences. Based on the clinical notes analyzed, this is due to the similarity of the information of physical and medical examinations and the summarized nature of the fillings out of the medical records by the doctors, which use many abbreviations and short sentences.

Table 3 shows an example of these characteristics: the patient's complaints (Subjective) come down to a short sentence with two nouns.

Table 3
Free-text clinical note example. The left column shows the original text in Portuguese with the sentence division. The column in the middle shows the translation. The column on the right shows the SOAP class.

Original (Portuguese)	Translation	SOAP
Reclama de cansaço e ronco.	Complaints about tiredness and snoring.	Subjective
Peso:115	Weight:115 (253.5 Pounds)	Objective
Altura:1,60	Height :1.60 (5.24 ft)	Objective
T.A:150 × 100	B.P.(blood pressure):150 × 100	Objective
Abdomen:globoso	Abdomen:globose	Objective
Mobilidade e força preservados.	Preserved mobility and strength.	Objective
Sobrepeso, possivelmente HAS.	Overweight, possibly SAH	Assessment
Oriento dieta.	Prescribed diet.	Plan
Solicito ECG e eco cardio.	Requested ECG and echo cardio.	Plan

Table 4
Sentences with most repetitions. The left column shows the original text in Portuguese. The column in the middle shows the translation. The column on the right shows the number of repetitions.

Original (Portuguese)	Translation	Qtd
Abdomen:SP	Abdomen:WP(Without Pathology)	31,893
Pele e mucosa:SP	Skin and mucosa:WP(Without Pathology)	29,017
T.A:120 × 80	B.P.(blood pressure):120 × 80	19,979
HAS	SAH (systemic arterial hypertension)	17,435
T.A:130 × 80	B.P.(blood pressure):130 × 80	11,675
T.A:140 × 80	B.P.(blood pressure):140 × 80	8519
DISLIPIDEMIA	DYSLIPIDEMIA	8005
Pele e mucosa:ÍNTEGRAS	Skin and mucosa:INTEGRAL	7879
FC:74	HR(heart rate):74	7789

The Objective is divided into five lines containing weight and height, which in Portuguese are measured in kilograms and meters, blood pressure (T.A), abdominal examination results, and physical description with little context. The Assessment is represented by a sentence indicating overweight and possible arterial hypertension represented by a medical abbreviation HAS/SAH. The Plan comes down to one sentence prescribing a diet and the second sentence requesting two medical tests in abbreviated forms Electrocardiogram (ECG/EKG) and Echocardiogram (echo cardio).

Table 4 shows the most repeated sentences, most of which result from routine physical examinations with average values. Of 2,979,987 sentences, 1,796,642 are duplicates resulting in 1,183,345 single sentences. For this study, the context of the sentence was not considered, as each sentence was classified only once, having the same classification for its other appearances.

To create a gold standard, we needed to label each sentence manually. Due to the limited number of researchers with authorization to access the database, a goal was to label approximately 100,000 single sentences (a GUI, Graphical User Interface, was developed to facilitate the labeling).

The labeling process was carried out as follows: an unlabeled medical record was randomly chosen, with the subsequent sentences labeled as Subjective, Objective, Assessment, and Plan or categorized as non-labelable. A new record was selected once a record's sentences were labeled. After the labeling of 101,645 sentences, all were reviewed. Three annotators carried out the entire process, each handling about a third of the sentences and reviewing another third without intersection. There were discrepancies in 4237 (4.16%) cases, which all three annotators then reviewed. The whole process took about 7 months.

Upon completion of this procedure, a total of 100,021 unique sentences were annotated with labels in the SOAP standard, and another 1624 were classified as non-labelable. The non-labelable sentences

Table 5
Distribution of manually labeled (gold standard) sentences containing the class, the number of sentences, and the proportion in relation to the total.

Label	Sentences	%
Subjective	48 784	48.77%
Objective	34 596	34.59%
Assessment	2751	2.75%
Plan	13 890	13.89%
Total	100 021	

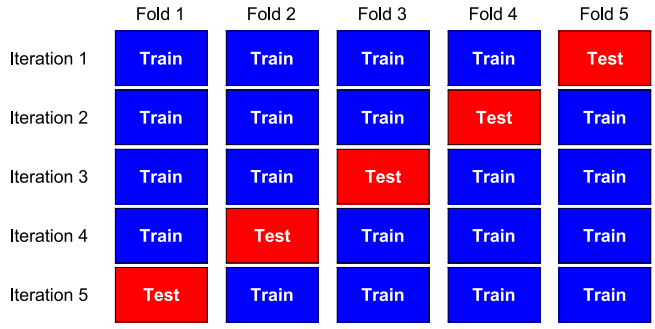


Fig. 2. Division of the k-fold into the 5 iterations.

mainly consisted of text without alphabetic characters, that presented only symbols such as "———", "*********", "!!!!!!!!", or sentences containing only the patient's name that were replaced by "#" and a random number in the anonymization process (example: #152 654). Only 210 of these sentences contained alphabetic characters. Due to the limited sample of these cases and the possibility of anonymization influencing the results in real-world situations, the focus was directed towards evaluating the performance of the models only with the sentences labeled in the SOAP standard.

In total, 15,883 clinical notes were labeled. Table 5 shows the distribution between the classes of single sentences. The low amount of Assessment is due to a lack of information about medical problems, diagnosis, and sentence repetition. Of 15,883 clinical notes, only 4791 had any Assessment, and 2040 were duplicates, resulting in the final 2751.

4.3. BioBERTptRT training

To create the BioBERTptRT, we retrained the BioBERTpt model using our private unlabeled dataset of 1,183,345 unique sentences. The sentences were tokenized using the BioBERTpt tokenizer. The model was trained for fifteen epochs, the same number used in the original training of BioBERTpt. Hyperparameters were configured with a batch size of 4, a learning rate of 2e−5, and a block size of 512. The training was conducted on a Quadro RTX 6000 24 GB GPU, using the Transformers PyTorch library developed by Hugging Face.[1]

4.4. Stratified k-fold and training

The stratified k-fold (k = 5) cross-validation method was applied to train and collect evaluation metrics. Stratified k-fold involves dividing the dataset into k subsets stratified, meaning that each subset's proportion of classes is preserved. Table 6 shows the division per class. Then, the model is trained and evaluated k times. Each iteration uses a different subset as the test set, while the remaining k − 1 subsets are used as the training set. Fig. 2 exemplifies this process. This approach ensures that each subset is used as the test set once, guaranteeing that all samples are used for both training and testing.

[1] https://github.com/huggingface/transformers.

Table 6
Distribution of the dataset between 5 folds.

Class	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5
Subjective	9757	9757	9757	9757	9756
Objective	6919	6919	6919	6919	6920
Assessment	551	550	550	550	550
Plan	2778	2778	2778	2778	2778
Total	20 005	20 004	20 004	20 004	20 004

The activation function employed was the multinomial logistic regression (Softmax). The Softmax function is utilized in multiclass models to compute the probabilities for each class, with the target class having the highest probability (Nwankpa, Ijomah, Gachagan, & Marshall, 2018).

All models were trained with the same fold divisions using the BERT implementation of PyTorch proposed by Hugging Face running in a GeForce, RTX 3070 SG, 8 GB, GDDR6 GPU with five epochs and hyperparameters set as maximum token size 512, batch size 32, and learning rate 2e−5. Experiments were conducted using more epochs, but no substantial gains or losses were observed.

4.5. Evaluation metrics

The following performance metrics were considered: accuracy, precision, recall, and F1-score, calculated individually for each class. As metrics, accuracy was calculated using raw and balanced, for precision, recall, and F1-score using weighted and macro. Macro average computes the metric independently for each class and then takes the average. This metric treats all classes equally and does not take imbalance into account. A weighted average takes the metrics from each class just like a macro, but the contribution of each class to the average is weighted by the number of examples available. Some data points contribute more than others (Chakravarthi, Muralidaran, Priyadharshini, & McCrae, 2020; Forman et al., 2003; Thorat, Tongaonkar, & Jagtap, 2020). The two metrics were calculated due to the Assessment class. As previously described, it has the lowest proportion among the classes and, as it will be presented in Section 5, the worst performance. However, it has no difference or prominence from other classes. An individual analysis per class was also conducted in the model with the highest performance, generating the confusion matrix, a precision–recall curve for each class, and the ROC curve. The ROC curve uses the one-versus-rest (OVR) approach. This approach is based on the idea that a classification system with c classes (c > 2) can also be interpreted as running c binary classifiers in parallel (Gimeno, Mingote, Ortega, Miguel, & Lleida, 2021).

5. Results and discussion

The evaluation results of each fold in its test interaction were used for computing the evaluation metrics. Table 7 summarizes all the results obtained from the validation of five models, exhibiting for each model true positive, true negative, false positive, false negative accuracy, precision, recall, and F1-score of each class and the total of each model. For accuracy, the raw and balanced total are exhibited. The weighted and macro are exhibited for precision, recall, and F1-score. The values in bold represent the highest result in the category among the models.

BioBERTptRT (F1-Score 94.36%) and BioBERTpt (F1-Score 94.08%) achieved the best results in all evaluated metrics. BioBERTpt demonstrated superior performance in precision and recall for the Plan class, while BioBERTptRT outperformed in all other metrics. However, the mBERT model performed the worst, with an F1-Score of 93.50%. The largest difference between classes was observed in the Assessment class, with a 4.28% increase in F1-Score when comparing mBERT (68.94%) to BioBERTptRT (73.23%). Fig. 3 shows a comparative graph of F1-Score with each class and model evaluated.

Table 7
Performance results for all models and classes. The left column shows the model with the results of each class and a totalizer line showing one value for True Positive, True Negative, False Positive, and False Negative, and two values for Accuracy (Raw/Balanced), Precision (Weighted/Macro), Recall (Weighted/Macro) and F1-score (Weighted/Macro).

Model/Class	True positive	True negative	False positive	False negative	Accuracy raw/ Balanced	Precision weighted/ Macro	Recall weighted/ Macro	F1-score weighted/ Macro
mBERT								
Subjective	45 849	48 389	2848	**2935**	0.9398	0.9415	0.9398	0.9406
Objective	33 022	63 601	**1824**	1574	0.9545	0.9476	0.9545	0.9510
Assessment	1793	96 613	**657**	**958**	0.6517	0.7318	0.6517	0.6894
Plan	12 903	85 006	**1125**	987	0.9289	0.9198	0.9289	0.9243
Total (Raw/Weighted)	93 567	293 609	**6454**	**6454**	0.9354	0.9348	0.9354	0.9350
Total (Balanced/Macro)					0.8687	0.8852	0.8687	0.8763
DistilBERT								
Subjective	46 034	48 418	2819	2750	0.9436	0.9422	0.9436	0.9429
Objective	33 017	63 689	1736	1579	0.9543	0.9500	0.9543	0.9521
Assessment	1883	96 624	646	868	0.6844	0.7445	0.6844	0.7132
Plan	12 864	85 109	1022	**1026**	0.9261	0.9264	0.9261	0.9262
Total (Raw/Weighted)	93 798	293 840	6223	6223	0.9377	0.9373	0.9377	0.9375
Total (Balanced/Macro)					0.8771	0.8908	0.8771	0.8836
BioBERT								
Subjective	46 127	48 076	**3161**	2657	0.9455	0.9358	0.9455	0.9406
Objective	32 744	63 846	1579	**1852**	0.9464	0.9539	0.9464	0.9502
Assessment	1880	96 634	636	871	0.6833	0.7472	0.6833	0.7138
Plan	12 869	85 106	1025	1021	0.9264	0.9262	0.9264	0.9263
Total (Raw/Weighted)	93 620	293 662	6401	6401	0.9360	0.9356	0.9360	0.9357
Total (Balanced/Macro)					0.8754	0.8908	0.8754	0.8827
BioBERTpt								
Subjective	46 057	48 646	2591	2727	0.9441	0.9467	0.9441	0.9454
Objective	33 060	63 845	1580	1536	0.9556	0.9543	0.9556	0.9549
Assessment	1893	**96 663**	607	858	0.6881	0.7572	0.6881	0.7210
Plan	**13 117**	85 015	1116	773	**0.9443**	0.9215	**0.9443**	0.9328
Total (Raw/Weighted)	94 127	294 169	5894	5894	0.9410	0.9406	0.9410	0.9408
Total (Balanced/Macro)					0.8830	0.8949	0.8830	0.8885
BioBERTptRT								
Subjective	**46 254**	**48 664**	2573	2530	**0.9481**	**0.9473**	**0.9481**	**0.9477**
Objective	**33 165**	**63 853**	1572	1431	**0.9586**	**0.9547**	**0.9586**	**0.9566**
Assessment	**1945**	96 654	616	806	**0.7070**	**0.7594**	**0.7070**	**0.7323**
Plan	13 034	**85 269**	862	856	0.9383	**0.9379**	0.9383	**0.9381**
Total (Raw/Weighted)	**94 398**	**294 440**	5623	5623	**0.9437**	**0.9434**	**0.9437**	**0.9435**
Total (Balanced/Macro)					**0.8880**	**0.8998**	**0.8880**	**0.8937**

Table 8
The F1-Score was compared in pairs of models under the Wilcoxon rank sum test, with bold values indicating statistical significance ($p < 0.05$).

p	BioBERTptRT	BioBERTpt	BioBERT	DistilBERT	mBERT
BioBERTptRT	–	**0.0317**	**0.0079**	**0.0079**	**0.0079**
BioBERTpt	**0.0317**	–	**0.0159**	**0.0159**	**0.0079**
BioBERT	**0.0079**	**0.0159**	–	0.1508	0.6905
DistilBERT	**0.0079**	**0.0159**	0.1508	–	0.0556
mBERT	**0.0079**	**0.0079**	0.6905	0.0556	–

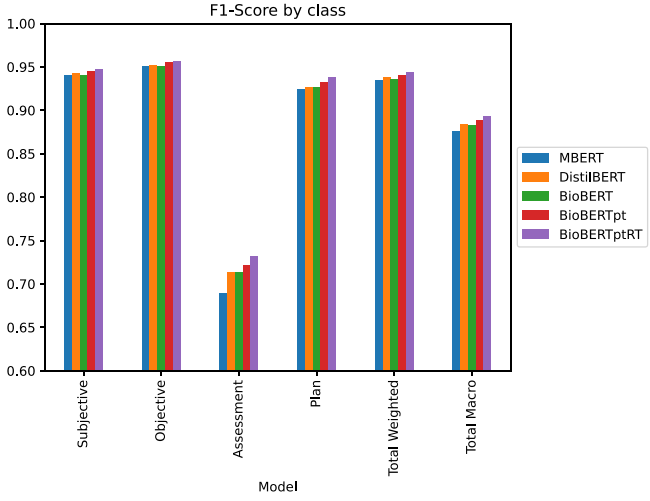


Fig. 3. F1-score graph of the four classes and five models (scale beginning in 0.6).

In order to measure whether the performance improvement in the F1-Score is statistically significant, the Wilcoxon rank sum test (Wilcoxon, Katti, Wilcox, et al., 1970) was utilized on the F1-Score results from each k-fold. The Wilcoxon rank sum test is a non-parametric method that determines whether the means are statistically different from each other based on ranks rather than the original F1-Scores of the experiments, being used to compare NPL models (Li et al., 2022). The results are displayed in Table 8, where only BioBERTptRT and BioBERTpt show a statistically significant gain compared to all the other models.

5.1. Analysis of the best model (BioBERTptRT)

Analyzing each class in the model individually, the Subjective, Objective, and Plan classes present similar results, with Objective having the best F1-score of 95.67%. The Assessment class presents the worst result. In Fig. 4, the confusion matrix displays the actual and predicted values for the four classes. The cells along the diagonal correspond to the true positive values for each class. The false negative values are shown in the rows outside the diagonal, while the false positive values are displayed in the columns outside the diagonal.

The most common type of confusion occurs between Subjective and Objective, resulting in a total of 2549 false positives. However, it should

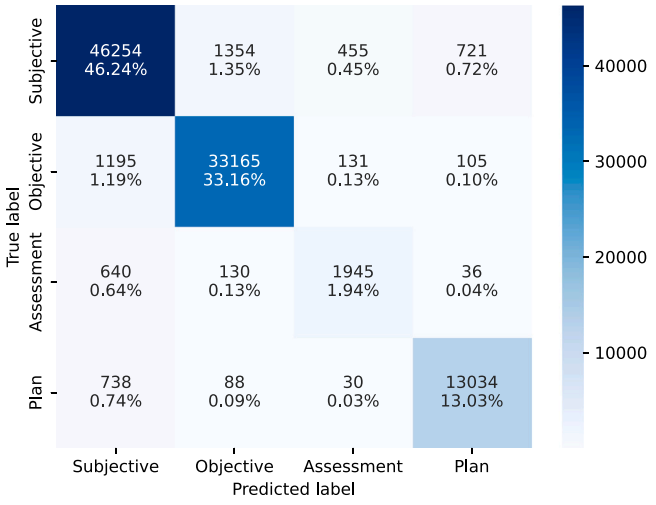


Fig. 4. BioBERTptRT confusion matrix with the relation of the true label and predicted label of the classes with the numerical value and percentage in relation to the total.

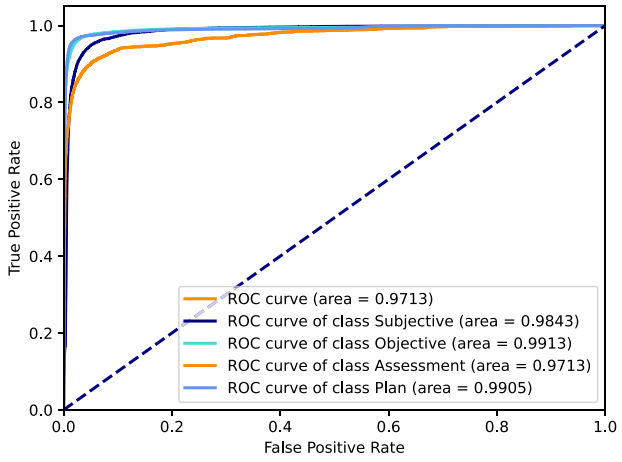


Fig. 5. BioBERTptRT - ROC Curve, the plot of test sensitivity (y coordinate) versus its false positive rate (x coordinate) of each class.

be noted that these two classes make up the majority of the dataset (83.36%). Fig. 5 shows (Receiver operating characteristic) curve of all classes using the one-versus-rest approach. Analyzing the curves and the area under a ROC curve (AUC) of the classes, which varied between 0.9713 and 0.9913, it is possible to affirm that the model presented an excellent performance in predicting a sentence concerning the SOAP class. Considering the dataset unbalancing, a precision–recall curve and its area for each class were also generated to make a more accurate analysis of the model's prediction quality. Fig. 6 shows this information. The classes of Subjective (area = 0.9787), Objective (area = 0.9864), and Plan (area = 0.9636) showed comparably high AUC values, while the Assessment class (area = 0.7655) exhibited a lower value. This is because both precision and recall measures are lower in the Assessment class than in the other classes.

5.2. The lower performance in the assessment class

Assessment is the class that presents the worst performance in all metrics when analyzing the dataset, the confusion matrix, and a sample analysis of false positives and false negatives resulting from the class. Two leading causes can be listed: the low sampling of the class compared to the others, which can cause a worse performance in the class in multiclass problems with an imbalanced dataset (Aldraimli et al., 2021; Sreejith, Khanna Nehemiah, & Kannan, 2020; Wang, Chen, Lin and Huang, 2021) and the difficulty of evaluating certain sentences without analyzing the context from the clinical note. It occurs mainly in false positives between the Assessment and the Subject. For example, the sentence "elevated HBP due to acute prostatitis" was evaluated as Assessment. However, in the context of the clinical note: "Patient was referred from a general internist.{new line}Elevated HBP due to acute prostatitis" is a subjective sentence because it represents a complaint brought by the referred patient. Without the sentence analysis in the clinical note context, it becomes ambiguous, making it difficult to classify the model. These cases were also identified in other classes, but it seems more recurrent in the Assessment class.

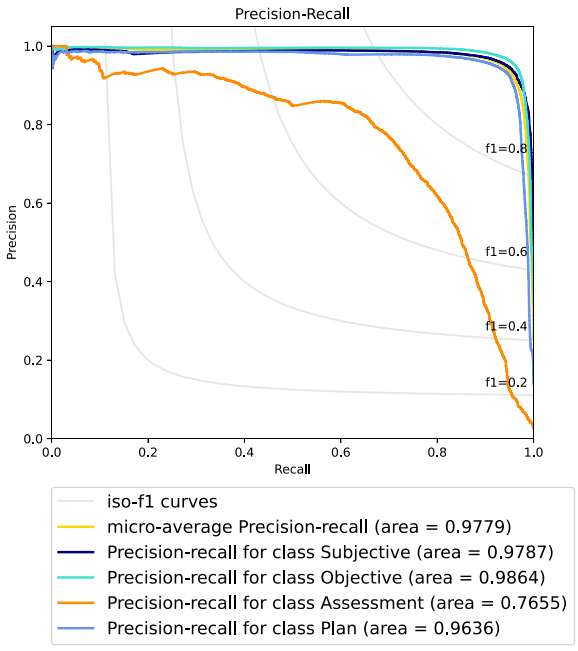


Fig. 6. BioBERTptRT Precision–Recall curve shows all classes' tradeoffs between precision and recall.

Table 9
Comparison of sentence distribution of our dataset and the dataset of Mowery et al. (2012).

Label	Mowery et al. (2012)			Our work		
	Sentences	%	F1-score	Sentences	%	F1-score
Subjective	1468	35.5%	0.939	48 784	48.77%	0.9477
Objective	1818	44.0%	0.945	34 596	34.59%	0.9566
Assessment	227	5.5%	0.757	2751	2.75%	0.7323
Plan	465	11.3%	0.770	13 890	13.89%	0.9435
Not applicable	335	8.1%				
Total	4130			100 021		

5.3. Discussion

Although not possible to compare our work to Mowery et al. (2012) directly, since they use different datasets and languages, it is possible to identify some similarities. Table 9 shows the amount and percentage of sentences in the dataset and the best F1-score of each class of each work. The work of Mowery et al. (2012) used a multilabel approach to model the problem, but only 165 (4.0%) sentences have two classes, and 9 (0.02%) have three classes. Both datasets present some similarities where the Subjective and Objective have the highest number of sentences, with the Objective higher in one dataset and the Subjective higher in another. The Assessment presents the lowest amount in both datasets, indicating that a lack of information about

Table 10
Comparison of the gain of BioBERTpt in relation to mBERT of our model and the models of Schneider et al. (2020) and Souza et al. (2021).

	Schneider et al. (2020)			Souza et al. (2021)			Proposed model		
	Precision	Recall	F1-score	Precision	Recall	F1-score	Precision	Recall	F1-score
mBERT	0.912	0.931	0.921	0.518	0.538	0.527	0.9349	0.9355	0.9335
BioBERTpt	0.917	0.935	0.926	0.566	0.557	0.561	0.9406	0.941	0.9408
Gain	0.005	0.004	0.005	**0.048**	**0.019**	**0.034**	0.006	0.005	0.007

Table 11
Ranking of models against the best weighted F1-Score and their domain specializations.

Model	Portuguese domain	Biomedical domain	Local domain	F1-score weighted	Position
BioBERTptRT	X	X	X	0.9435	1º
BioBERTpt	X	X		0.9408	2º
DistilBERT	X			0.9375	3º
BioBERT		X		0.9357	4º
mBERT	X			0.9350	5º

medical problems diagnosis shown in Section 4.2 is recurrent in other datasets. The Assessment presents the worst performance in both. This could imply a higher level of complexity in classifying the class discussed in the preceding paragraph. Alternatively, it could suggest that insufficient sentences are available to train the network adequately. This is particularly evident in the Plan class, where our work achieved favorable results and possessed more sentences in the dataset.

5.4. BioBERTpt specialization

Through domain specialization, it was possible to identify a gain in the proposed task. Table 11 presents the model's classification concerning the best F1-Score (weighted). BioBERTptRT achieved the best result and is also the most specialized, with three domains (Portuguese, biomedical, and local), followed by BioBERTpt with two specializations (Portuguese and biomedical). With only one specialization, the other models occupy the last three positions.

It is relevant to note that DistilBERT performed better than mBERT, despite having the same domain. One possible explanation is that mBERT is multilingual, while the DistilBERT used in the study was trained exclusively for Portuguese. Analyzing the result of BioBERT, it can be concluded that the biomedical domain alone only brings significant gains with the language domain.

Specializing for medical groups or hospital networks resulted in a 0.28% increase in F1 score compared to BioBERTpt without this specialization. Despite this real gain, it is important to consider some issues when assessing the feasibility of this process. Pre-training requires a substantially larger dataset compared to fine-tuning. For pre-training, 1,183,345 unlabeled sentences were used, compared to 100,021 labeled sentences for fine-tuning. Additionally, it was not possible to perform the pre-training using the same hardware, requiring more advanced computational resources, such as a 24 GB Quadro RTX 6000 GPU, with a maximum batch size of 4, in contrast to the fine-tuning performed on an 8 GB RTX 3070 GPU, with a batch size of 32. Furthermore, this model is expected to perform less when tested on datasets from other medical groups or hospital networks.

5.5. Comparison of domain specialization in other works

In other studies, it has been identified that domain specialization, in addition to Portuguese language specialization, leads to gains. Although this improvement is small, it remains consistent when we compare our work with the works proposed by Schneider et al. (2020) and Souza et al. (2021). These studies also compare mBERT and BioBERTpt but used different datasets and objectives. The obtained results are similar.

Table 10 presents the performance gain in terms of precision, recall, and F1-score in the three works. The study conducted by Souza et al. (2021) shows a larger gain in the F1-score (0.034), but the absolute value of the F1-score is only 0.561. On the other hand, both the model proposed in this study and Schneider et al.'s (2020) model shows smaller gains in F1-score (0.004 and 0.005, respectively), but their absolute F1-score values are higher (0.9461 and 0.926), which may explain this smaller difference.

Based on this analysis, we can affirm that domain specialization in the clinical field demonstrates superior performance in NLP tasks in medical domains in Portuguese compared to models that only specialize in Portuguese.

6. Conclusion

This study proposed a model for classifying sentences in clinical notes from electronic medical records in Portuguese, following the SOAP note standard, using domain-specific pre-trained language models. Among the five pre-trained BERT models tested, BioBERTptRT achieved the best results. It achieved an accuracy of 0.9461, a precision of 0.9434, a recall of 0.9437, and an F1-score of 0.9435. BioBERTptRT is the most specialized model, possessing specific knowledge of the Portuguese language, clinical terminology, and the medical group's context. Compared to the second most specialized model, BioBERTptRT obtained a 0.28% increase in the F1-score.

In the individual analysis by class of the best model, the class that exhibited the poorest performance was Assessment. It achieved an accuracy of 0.7070, precision of 0.7594, recall of 0.7070, and an F1-score of 0.7323. This outcome can be attributed to two primary factors: firstly, the limited number of sentences compared to other classes, and secondly, the absence of contextual information from other sentences in the medical record. Notably, this class demonstrated the highest improvement (4.28%) with domain specialization.

As future work to improve the results of the Assessment class, we propose increasing the dataset of the class to verify if it increases the performance. Another approach contextualizes the sentences using BERT's 'SEP' special separator token to include together with the current sentence, the previous one, or the prediction already made of this previous sentence.

As the project evolves, the next step is to increase the degree of structuring of the information. It can be achieved by structuring the information in classified sentences. Using medical-named entity recognition techniques (MER/MNER) (Soriano & Castro, 2017) and medical relation extraction (MRE) (Fu, Li, & Ma, 2019) to structure the sentences in ontologies (Slater et al., 2021) or SNOMED CT Expressions (Donnelly, 2006).

The source code for the work is available on GitHub,[2] but since the dataset used in the work is private, we cannot make it available.

CRediT authorship contribution statement

Jezer Machado de Oliveira: Writing – original draft, Conceptualization, Methodology, Investigation, Software. **Rodolfo Stoffel Antunes:** Writing – review & editing, Conceptualization, Methodology, Investigation. **Cristiano Andrẽ da Costa:** Writing – review & editing, Conceptualization, Methodology, Investigation, Supervision.

[2] https://github.com/jezer-oliveira/BertSOAPClassifier

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Cristiano Andre da Costa reports financial support was provided by Coordination of Higher Education Personnel Improvement and National Council for Scientific and Technological Development. Jezer Machado de Oliveira reports financial support was provided by instituto Federal de Educação, Ciência e Tecnologia do Rio Grande do Sul.

Data availability

The authors do not have permission to share data.

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References

Aldraimli, M., Soria, D., Grishchuck, D., Ingram, S., Lyon, R., Mistry, A., et al. (2021). A data science approach for early-stage prediction of Patient's susceptibility to acute side effects of advanced radiotherapy. *Computers in Biology and Medicine*, *135*, Article 104624. http://dx.doi.org/10.1016/j.compbiomed.2021.104624, URL https://www.sciencedirect.com/science/article/pii/S0010482521004182.

Almeida, J. R., & Matos, S. (2020). Rule-based extraction of family history information from clinical notes. In *Proceedings of the 35th annual ACM symposium on applied computing*. ACM, http://dx.doi.org/10.1145/3341105.3374000.

Alsentzer, E., Murphy, J. R., Boag, W., Weng, W.-H., Jin, D., Naumann, T., et al. (2019). Publicly available clinical BERT embeddings. arXiv:1904.03323.

Alvarez, E. E., & Reinhart, J. M. (2020). Use of an interactive online teaching module improved students' ability to write a clinically appropriate SOAP note. *J. Vet. Med. Educ.*, *47*(6), 700–708.

Ammar, W., Groeneveld, D., Bhagavatula, C., Beltagy, I., Crawford, M., Downey, D., et al. (2018). Construction of the literature graph in semantic scholar. arXiv preprint arXiv:1805.02262.

Beltagy, I., Lo, K., & Cohan, A. (2019). SciBERT: A pretrained language model for scientific text. arXiv preprint arXiv:1903.10676.

Cameron, S., & Turtle-Song, I. (2002). Learning to write case notes using the SOAP format. *J. Counsel. Dev.*, *80*(3), 286–292.

Cegalla, D. P. (2008). *Novíssima gramática da língua Portuguesa*. Companhia Ed. Nacional.

Chakravarthi, B. R., Muralidaran, V., Priyadharshini, R., & McCrae, J. P. (2020). Corpus creation for sentiment analysis in code-mixed Tamil-English text. arXiv preprint arXiv:2006.00206.

Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 conference of the north american chapter of the association for computational linguistics: human language technologies, Vol. 1 (Long and Short Papers)* (pp. 4171–4186). Minneapolis, Minnesota: Association for Computational Linguistics, http://dx.doi.org/10.18653/v1/N19-1423, URL https://www.aclweb.org/anthology/N19-1423.

Donnelly, K. (2006). SNOMED-CT: The advanced terminology and coding system for eHealth. *Studies in Health Technology and Informatics*, *121*, 279.

Forman, G., et al. (2003). An extensive empirical study of feature selection metrics for text classification.. *Journal of Machine Learning Research*, *3*(Mar), 1289–1305.

Fu, S., Chen, D., He, H., Liu, S., Moon, S., Peterson, K. J., et al. (2020). Clinical concept extraction: a methodology review. *Journal of Biomedical Informatics*, Article 103526.

Fu, T.-J., Li, P.-H., & Ma, W.-Y. (2019). Graphrel: Modeling text as relational graphs for joint entity and relation extraction. In *Proceedings of the 57th annual meeting of the association for computational linguistics* (pp. 1409–1418).

Gimeno, P., Mingote, V., Ortega, A., Miguel, A., & Lleida, d. (2021). Generalizing AUC optimization to multiclass classification for audio segmentation with limited training data. *IEEE Signal Processing Letters*, *28*, 1135–1139. http://dx.doi.org/10.1109/LSP.2021.3084501.

Ilan, R., LeBaron, C. D., Christianson, M. K., Heyland, D. K., Day, A., & Cohen, M. D. (2012). Handover patterns: an observational study of critical care physicians. *BMC Health Serv. Res.*, *12*(1), 1–10.

Kenei, J. K., Moso, J. C., Omullo, E. T. O., & Oboko, R. (2019). Deep CNN with residual connections and range normalization for clinical text classification. *Computer Science and Information Technology*, *7*(4), 111–127. http://dx.doi.org/10.13189/csit.2019.070402.

Kersloot, M. G., van Putten, F. J., Abu-Hanna, A., Cornet, R., & Arts, D. L. (2020). Natural language processing algorithms for mapping clinical text fragments onto ontology concepts: a systematic review and recommendations for future studies. *Journal of Biomedical Semantics*, *11*(1), 1–21.

Kim, M. C., Nam, S., Wang, F., & Zhu, Y. (2020). Mapping scientific landscapes in UMLS research: a scientometric review. *Journal of the American Medical Informatics Association*, *27*(10), 1612–1624.

Lee, J., Yoon, W., Kim, S., Kim, D., Kim, S., So, C. H., et al. (2019). BioBERT: a pre-trained biomedical language representation model for biomedical text mining. *Bioinformatics*, *36*(4), 1234–1240. http://dx.doi.org/10.1093/bioinformatics/btz682, arXiv:https://academic.oup.com/bioinformatics/article-pdf/36/4/1234/32527770/btz682.pdf.

Li, J., Wei, Q., Ghiasvand, O., Chen, M., Lobanov, V., Weng, C., et al. (2022). A comparative study of pre-trained language models for named entity recognition in clinical trial eligibility criteria from multiple corpora. *BMC Medical Informatics and Decision Making*, *22*(3), 1–10.

Lisenby, K. M., Andrus, M. R., Jackson, C. W., Stevenson, T. L., Fan, S., Gaillard, P., et al. (2018). Ambulatory care preceptors' perceptions on SOAP note writing in advanced pharmacy practice experiences (APPEs). *Currents in Pharmacy Teaching and Learning*, *10*(12), 1574–1578. http://dx.doi.org/10.1016/j.cptl.2018.09.002, URL https://www.sciencedirect.com/science/article/pii/S1877129718300261.

Maximilian, Z., J., B. O., & Michael, M. (2019). Using openEHR archetypes for automated extraction of numerical information from clinical narratives. *Studies in Health Technology and Informatics*, *267*(German Medical Data Sciences: Shaping Change – Creative Solutions for Innovative Medicine), 156–163. http://dx.doi.org/10.3233/SHTI190820.

Moen, H., Hakala, K., Peltonen, L.-M., Matinolli, H.-M., Suhonen, H., Terho, K., et al. (2020). Assisting nurses in care documentation: from automated sentence classification to coherent document structures with subject headings. *Journal of Biomedical Semantics*, *11*(1), http://dx.doi.org/10.1186/s13326-020-00229-7.

Moen, H., Hakala, K., Peltonen, L.-M., Suhonen, H., Ginter, F., Salakoski, T., et al. (2019). Supporting the use of standardized nursing terminologies with automatic subject heading prediction: a comparison of sentence-level text classification methods. *Journal of the American Medical Informatics Association*, *27*(1), 81–88. http://dx.doi.org/10.1093/jamia/ocz150.

Moen, H., Hakala, K., Peltonen, L.-M., Suhonen, H., Loukasmäki, P., Salakoski, T., et al. (2018). Evaluation of a prototype system that automatically assigns subject headings to nursing narratives using recurrent neural network. In *Proceedings of the ninth international workshop on health text mining and information analysis*. Association for Computational Linguistics, http://dx.doi.org/10.18653/v1/w18-5611.

Mowery, D., Wiebe, J., Visweswaran, S., Harkema, H., & Chapman, W. W. (2012). Building an automated SOAP classifier for emergency department reports. *Journal of Biomedical Informatics*, *45*(1), 71–81. http://dx.doi.org/10.1016/j.jbi.2011.08.020, URL https://www.sciencedirect.com/science/article/pii/S153204641100147X.

Nwankpa, C., Ijomah, W., Gachagan, A., & Marshall, S. (2018). Activation functions: Comparison of trends in practice and research for deep learning. arXiv preprint arXiv:1811.03378.

de Oliveira, J. M., da Costa, C. A., & Antunes, R. S. (2021). Data structuring of electronic health records: a systematic review. *Health and Technology*, *11*(6), 1219–1235.

Qin, X., Liu, J., Wang, Y., Liu, Y., Deng, K., Ma, Y., et al. (2021). Natural language processing was effective in assisting rapid title and abstract screening when updating systematic reviews. *Journal of Clinical Epidemiology*.

Reys, A. D., Silva, D., Severo, D., Pedro, S., de Sousa e Sá, M. M., & Salgado, G. A. C. (2020). Predicting multiple ICD-10 codes from Brazilian-portuguese clinical notes. In R. Cerri, & R. C. Prati (Eds.), *Intelligent systems* (pp. 566–580). Cham: Springer International Publishing.

Roehrs, A., Da Costa, C. A., da Rosa Righi, R., & De Oliveira, K. S. F. (2017). Personal health records: a systematic literature review. *Journal of Medical Internet Research*, *19*(1), Article e13.

Rosenbloom, S. T., Denny, J. C., Xu, H., Lorenzi, N., Stead, W. W., & Johnson, K. B. (2011). Data from clinical notes: a perspective on the tension between structure and flexible documentation. *Journal of the American Medical Informatics Association*, *18*(2), 181–186. http://dx.doi.org/10.1136/jamia.2010.007237, arXiv:https://academic.oup.com/jamia/article-pdf/18/2/181/6035310/18-2-181.pdf.

Ross, S. E., Schilling, L. M., Fernald, D. H., Davidson, A. J., & West, D. R. (2010). Health information exchange in small-to-medium sized family medicine practices: Motivators, barriers, and potential facilitators of adoption. *International Journal of Medical Informatics*, *79*(2), 123–129. http://dx.doi.org/10.1016/j.ijmedinf.2009.12.001, URL https://www.sciencedirect.com/science/article/pii/S1386505609001798.

Sanh, V., Debut, L., Chaumond, J., & Wolf, T. (2019). DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter. arXiv preprint arXiv:1910.01108.

Santos, D., Dutra, F., Parreiras, F., & Brandão, W. (2021). Assessing the effectiveness of multilingual transformer-based text embeddings for named entity recognition in portuguese. In *Proceedings of the 23rd international conference on enterprise information systems - Volume 1: ICEIS* (pp. 473–483). SciTePress, INSTICC, http://dx.doi.org/10.5220/0010443204730483.

Schneider, E. T. R., de Souza, J. V. A., Knafou, J., e Oliveira, L. E. S., Copara, J., Gumiel, Y. B., et al. (2020). BioBERTpt-A portuguese neural language model for clinical named entity recognition. In *Proceedings of the 3rd clinical natural language processing workshop* (pp. 65–72).

Silva e Oliveira, L. E., Peters, A. C., da Silva, A. M. P., Gebeluca, C. P., Gumiel, Y. B., Cintho, L. M. M., et al. (2020). SemClinBr - a multi institutional and multi specialty semantically annotated corpus for portuguese clinical NLP tasks. CoRR abs/2001.10071 arXiv:2001.10071 URL https://arxiv.org/abs/2001.10071.

Slater, L. T., Williams, J. A., Karwath, A., Fanning, H., Ball, S., Schofield, P. N., et al. (2021). Multi-faceted semantic clustering with text-derived phenotypes. *Computers in Biology and Medicine*, *138*, Article 104904. http://dx.doi.org/10.1016/j.compbiomed.2021.104904, URL https://www.sciencedirect.com/science/article/pii/S0010482521006983.

Soriano, I. M., & Castro, J. (2017). DNER clinical (named entity recognition) from free clinical text to snomed-CT concept. *WSEAS Transactions on Computers*, *16*, 83–91.

Souza, J. V. A., Schneider, E. T. R., Cezar, J. O., Silva, L. E., Gumiel, Y. B., Paraiso, E. C., et al. (2021). A multilabel approach to portuguese clinical named entity recognition. *Journal of Health Informatics*, *12*.

Sreejith, S., Khanna Nehemiah, H., & Kannan, A. (2020). Clinical data classification using an enhanced SMOTE and chaotic evolutionary feature selection. *Computers in Biology and Medicine*, *126*, Article 103991. http://dx.doi.org/10.1016/j.compbiomed.2020.103991, URL https://www.sciencedirect.com/science/article/pii/S001048252030322X.

Symeonidou, A., Sazonau, V., & Groth, P. (2019). Transfer learning for biomedical named entity recognition with BioBERT. In *SEMANTICS posters & demos*.

Thorat, P., Tongaonkar, R., & Jagtap, V. (2020). Towards designing the best model for classification of fish species using deep neural networks. In S. Bhalla, P. Kwan, M. Bedekar, R. Phalnikar, & S. Sirsikar (Eds.), *Proceeding of international conference on computational science and applications* (pp. 343–351). Singapore: Springer Singapore.

Tognola, G., Murri, A., & Cuda, D. (2018). Cognitive computing for the automated extraction and meaningful use of health data in narrative medical notes: An application to the clinical management of hearing impaired aged patients. In *2018 IEEE EMBS international conference on biomedical & health informatics (BHI)*. IEEE, http://dx.doi.org/10.1109/bhi.2018.8333428.

Tomanek, K., Wermter, J., & Hahn, U. (2007). Sentence and token splitting based on conditional random fields. In *Proceedings of the 10th conference of the pacific association for computational linguistics, Vol. 49* (p. 57). Citeseer.

Wang, L., Chen, Z., Lin, N., & Huang, X. (2021). An interdisciplinary literature classifier based on multi-task multi-label learning. In *2021 international conference on asian language processing (IALP)* (pp. 183–188). http://dx.doi.org/10.1109/IALP54817.2021.9675234.

Wang, B., Hu, X., Li, P., & Yu, P. S. (2021). Cognitive structure learning model for hierarchical multi-label text classification. *Knowledge-Based Systems*, *218*, Article 106876. http://dx.doi.org/10.1016/j.knosys.2021.106876, URL https://www.sciencedirect.com/science/article/pii/S0950705121001398.

Wilcoxon, F., Katti, S., Wilcox, R. A., et al. (1970). Critical values and probability levels for the wilcoxon rank sum test and the Wilcoxon signed rank test. *Selected Tables in Mathematical Statistics*, *1*, 171–259.

Wu, S., & Dredze, M. (2020). Are all languages created equal in multilingual BERT?. (pp. 120–130). http://dx.doi.org/10.18653/v1/2020.repl4nlp-1.16.

Wu, P.-H., Yu, A., Tsai, C.-W., Koh, J.-L., Kuo, C.-C., & Chen, A. L. P. (2020). Keyword extraction and structuralization of medical reports. *Health Information Science and Systems*, *8*(1), http://dx.doi.org/10.1007/s13755-020-00108-6.

Xie, Q., Bishop, J. A., Tiwari, P., & Ananiadou, S. (2022). Pre-trained language models with domain knowledge for biomedical extractive summarization. *Knowledge-Based Systems*, *252*, Article 109460. http://dx.doi.org/10.1016/j.knosys.2022.109460, URL https://www.sciencedirect.com/science/article/pii/S0950705122007328.

Zhang, R., Chu, F., Chen, D., & Shang, X. (2018). A text structuring method for Chinese medical text based on temporal information. *International Journal of Environmental Research and Public Health*, *15*(3), http://dx.doi.org/10.3390/ijerph15030402, URL https://www.mdpi.com/1660-4601/15/3/402.